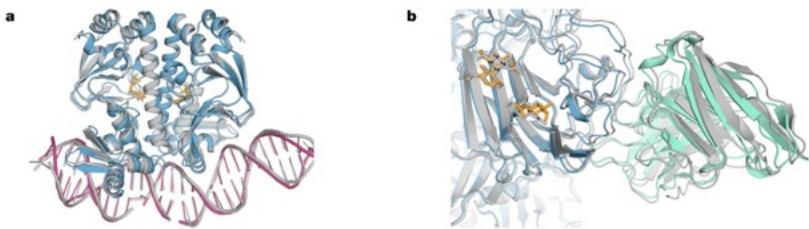


From Target Chemistry to AI-Assisted PD

- Use what you know about target structure and chemical recognition.
- Ask how AI helps choose molecules and the experiments that test them.

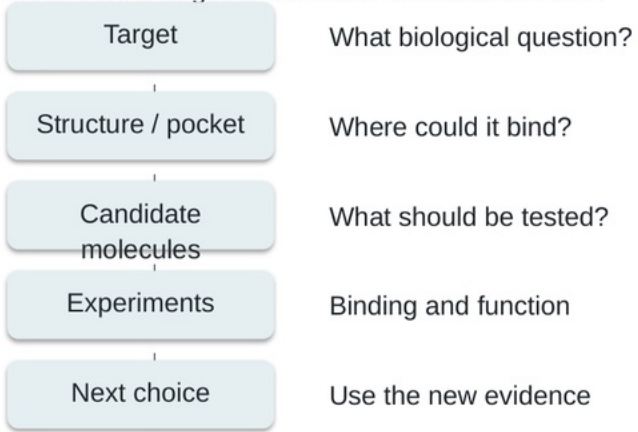


Published AF3 complexes: different molecules require different recognition models.

Abramson et al., Nature 630, 493–500 (2024).

Follow a Drug–Target Research Question

- Define the interaction or effect before choosing a tool.
- Move from a target to a testable chemical decision.



Target → structure → candidates → experiments → next decision.

Teaching workflow.

AI Tools for Pharmacodynamic Research

- Choose by the question, the available data and the meaning of the output.

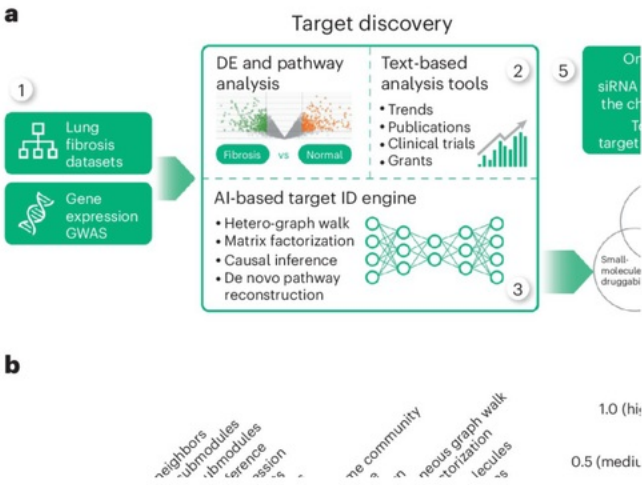
Research question	Example tool	Typical result
What could the complex look like?	AlphaFold 3	Predicted 3D arrangement
Where might a ligand bind?	P2Rank / PrankWeb	Candidate binding pockets
How might a ligand fit?	GNINA	Docked poses and scores
Which pairs look promising?	Boltz-2	Structure and binding outputs
Which analogue may be active?	Chemprop	Assay-trained predictions
What could we make next?	REINVENT 4	Candidate chemical structures

Examples of useful research tools—not a popularity or performance ranking.

Official documentation and original tool papers; details in notes.

Use Disease Evidence to Prioritize Targets

- Combine disease-related measurements with established biological knowledge.
- AI ranking proposes a target; it does not validate the target.

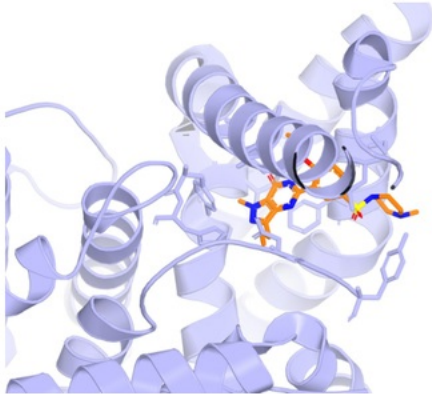


TNIK study: biological evidence used in target prioritization.

Ren et al., Nat. Biotechnol. 43, 63–75 (2025; online 2024).

Enzymes: Start from a Real Binding Complex

- Sildenafil–PDE5 provides a known inhibitor and a defined pocket.
- Use that reference to prepare an AI-assisted screening question.

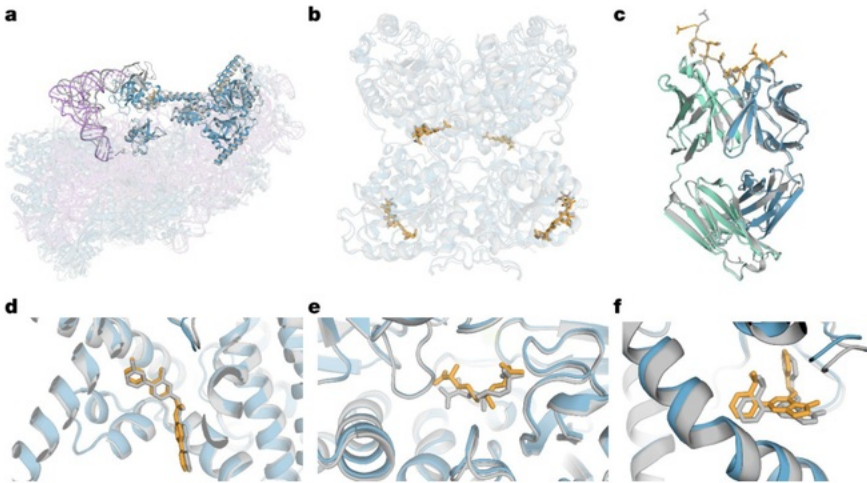


Experimental sildenafil–PDE5 structure, PDB 1UDT.

RCSB PDB 1UDT.

Predict a Complex When a Structure Is Missing

- Provide the identities of the interacting molecules.
- Use predicted geometry as a starting hypothesis—not measured activity.

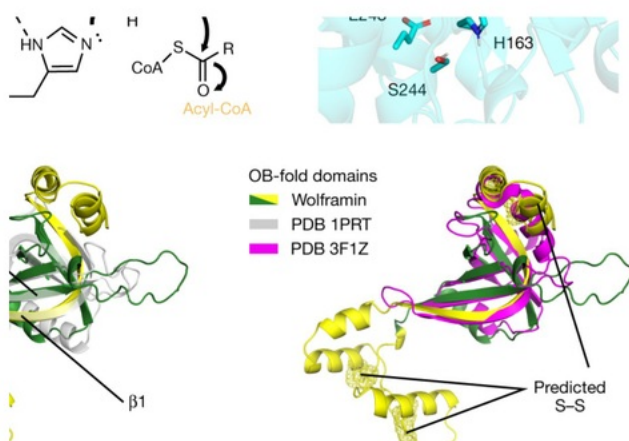


AF3 examples include protein–RNA, antibody and small-molecule complexes.

Abramson et al., Nature 630, 493–500 (2024).

Find Candidate Pockets with P2Rank

- A pocket predictor highlights regions that may bind a ligand.
- Known chemistry and experiments help decide which site matters.



Published DGAT2 pocket illustration; a candidate region is not a confirmed drug site.

Tunyasuvunakool et al., Nature 596, 590–596 (2021); P2Rank.

13

GPCRs: The Receptor State Matters

- Active and inactive receptors need not present the same geometry.
- Choose a relevant state before interpreting a ligand prediction.



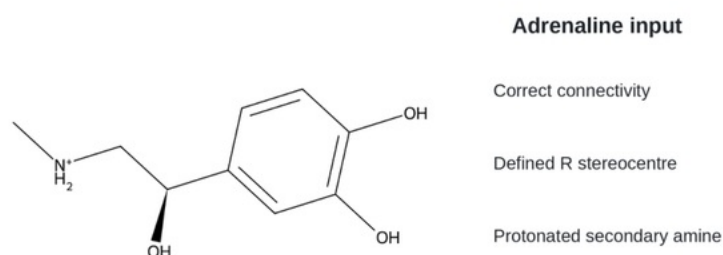
β₂-adrenoceptor: inactive and active experimental structures; TM6 highlighted.

RCSB PDB 2RH1 and 3SN6; Rasmussen et al., Nature 477, 549–555 (2011).

14

Give the Tool the Correct Chemical Structure

- Keep connectivity, charge and stereochemistry explicit.
- Adrenaline illustrates why those chemical details cannot be ignored.



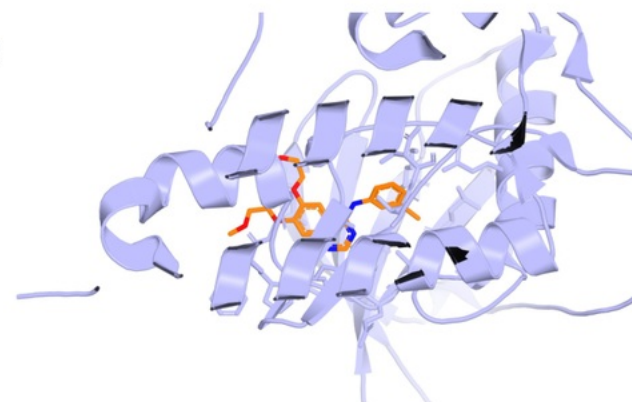
R-adrenaline shown with a protonated secondary amine.

Original chemical illustration; molecular identity checked computationally.

15

Docking: Propose How a Ligand Fits

- GNINA combines docking with learned pose scoring.
- Compare the proposal with known contacts and experimental structures.



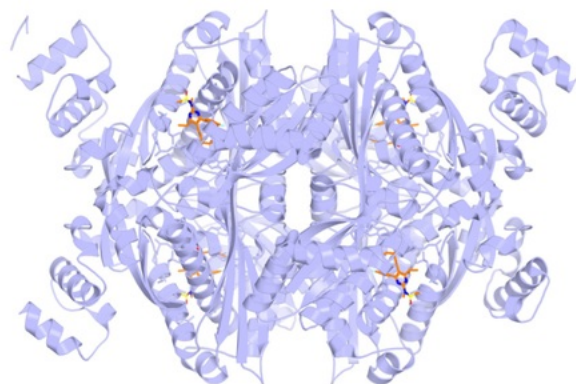
Experimental erlotinib–EGFR reference, PDB 1M17; not a new GNINA result.

McNutt et al., J. Cheminform. 17, 28 (2025). RCSB PDB 1M17.

16

Inspect Contacts with the Chemistry You Know

- Check whether polar and nonpolar groups face suitable environments.
- A contact picture suggests chemistry; contact counting does not measure affinity.



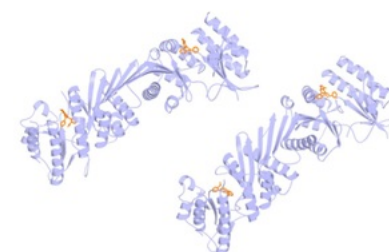
Rosuvastatin in the experimental HMG-CoA reductase complex, PDB 1HWL.

Istvan & Deisenhofer, Science 292, 1160–1164 (2001); PDB 1HWL.

17

Search Beyond the Usual Active Site

- An allosteric ligand can act at a different pocket.
- Specify the intended site when predicting or scoring binding.



Experimental allosteric inhibitor complex of PI5P4Ky, PDB 7QIE.

Abramson et al., Nature 630, 493–500 (2024). RCSB PDB 7QIE.

18

Boltz-2: Read the Right Output

- Structural confidence, binding likelihood and activity estimates are different.
- Validate the particular claim represented by each output.

Structure	Is this molecular arrangement plausible?
Binding-related prediction	What precisely was the model trained to predict?
Experimental result	Which binding or functional assay checks it?

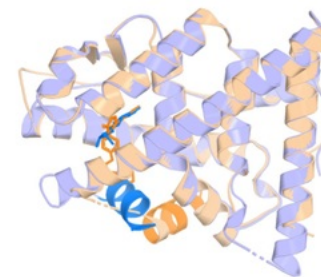
A confident-looking structure is not an experimental binding measurement.

Boltz-2 original paper and official prediction documentation (2025).

19

Nuclear Receptors: Binding Is Not Function

- Oestradiol and raloxifene bind the receptor but stabilize different arrangements.
- A pose prediction alone cannot determine agonism or antagonism.



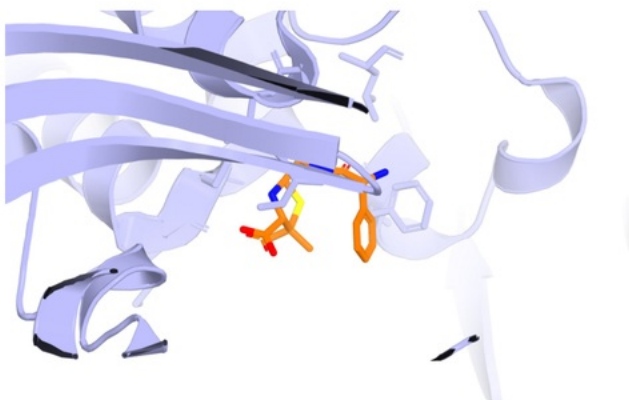
Oestrogen receptor comparison: oestradiol blue; raloxifene wheat/orange.

Brzozowski et al., Nature 389, 753–758 (1997); PDB 1ERE/1ERR.

20

Covalent Inhibitors Need a Covalent-Aware Model

- Ampicillin forms a covalent acyl-enzyme complex with a PBP.
- Check the reaction and the bound chemical form—not just the fit.



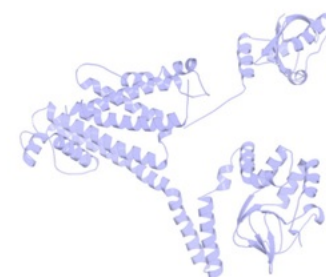
Experimental covalent ampicillin–PBP4 complex, PDB 2EX6.

Kishida et al., Biochemistry 45, 783–792 (2006). McNutt et al., J. Cheminform. 17, 28 (2025).

21

Ion Channels: Flag Possible hERG Blockers

- HERGAI predicts hERG inhibition from molecule–channel information.
- Use the prediction to prioritize electrophysiology tests.



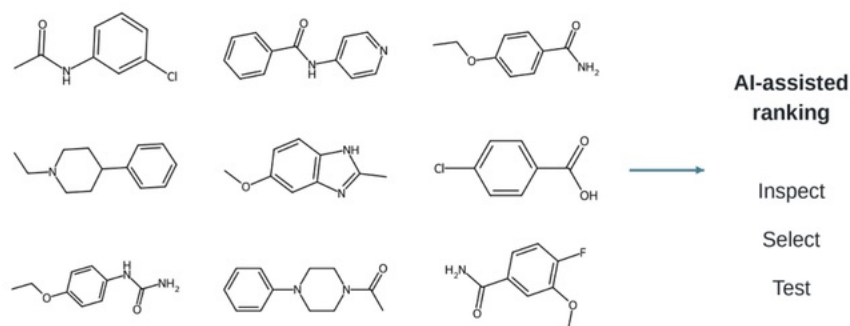
Experimental hERG channel structure, PDB 5VA1; no blocker pose is asserted.

Tran-Nguyen et al., J. Cheminform. 17, 110 (2025); PDB 5VA1.

22

Screen Broadly; Test a Manageable Shortlist

- AI-assisted ranking can reduce the number of compounds sent to the laboratory.
- Keep chemically sensible and sufficiently diverse candidates.



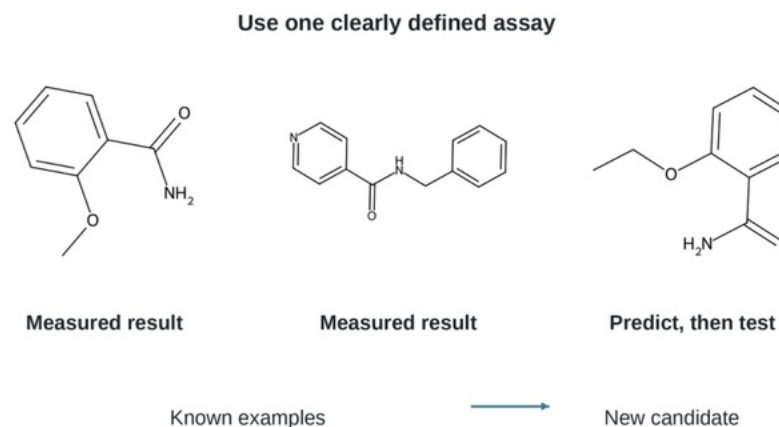
Illustrative candidate collection; no activities or model scores are assigned.

McNutt et al., J. Cheminform. 17, 28 (2025).

23

Predict Activity from Measured Examples

- Chemprop can learn a defined activity endpoint from appropriate assay data.
- Test predictions on compounds not used to build the model.



A structure and an assay result form a learning example.

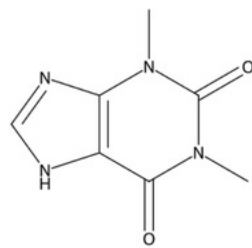
Heid et al., J. Chem. Inf. Model. 64, 9–17 (2024).

24

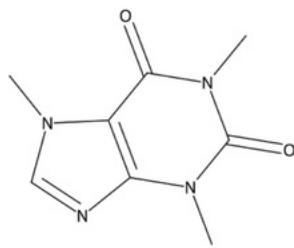
Small Chemical Changes Can Matter

- Theophylline and caffeine differ by one methyl group.
- A model must capture the recognition problem—not just overall similarity.

A single methyl group changes the recognition problem



Theophylline



Caffeine

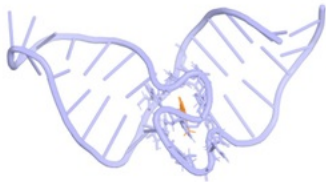
A theophylline-binding RNA aptamer discriminates against caffeine.

Zimmermann et al., Nat. Struct. Biol. 4, 644–649 (1997).

25

RNA Targets Need RNA-Relevant Models

- SMRTnet combines RNA sequence/secondary structure with the candidate molecule.
- Protein-trained binding predictions should not be transferred to RNA without testing.



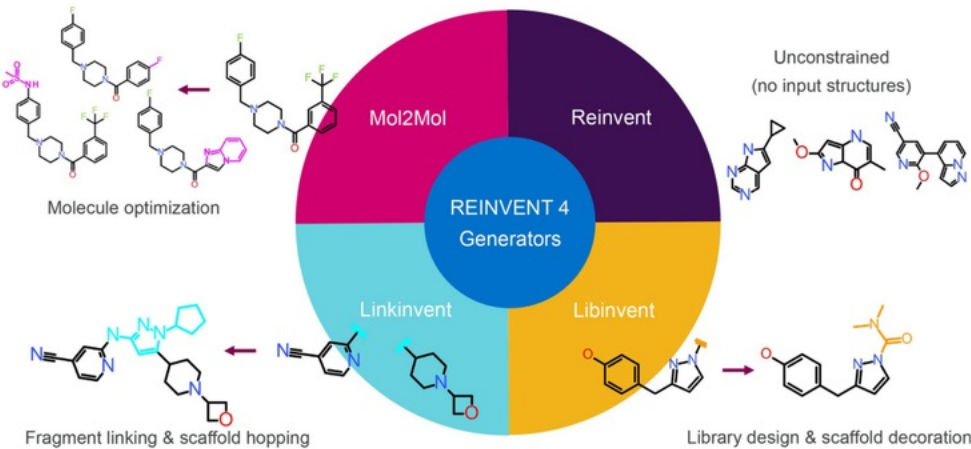
Experimental theophylline–RNA aptamer, PDB 1EHT; one NMR model.

Fei et al., Nat. Biotechnol. (2026), doi:10.1038/s41587-025-02942-z. PDB 1EHT.

26

Ask AI to Propose New Chemical Ideas

- REINVENT 4 can decorate a scaffold, link fragments or propose new molecules.
- Design goals guide proposals; synthesis and assays determine their value.



Four molecular-design tasks in REINVENT 4.

Loeffler et al., J. Cheminform. 16, 20 (2024).

27

Predict a Target Profile, Not Just One Score

- Related proteins and unwanted targets may also interact with the molecule.
- Follow promising predictions with an appropriate target panel.

	Target A	Target B	Target C	hERG
Candidate 1				
Candidate 2				
Candidate 3				

Darker cells = stronger inhibition in this invented example

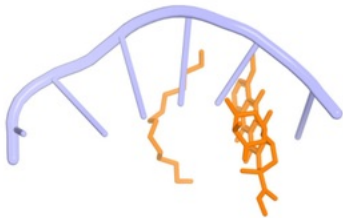
Hypothetical matched-assay inhibition patterns—not measured candidates.

Hopkins, Nat. Chem. Biol. 4, 682–690 (2008).

28

DNA Recognition Is a Different Binding Problem

- Intercalation and groove recognition require the appropriate molecular geometry.
- Use DNA-relevant data and validation, not untested transfer from protein models.



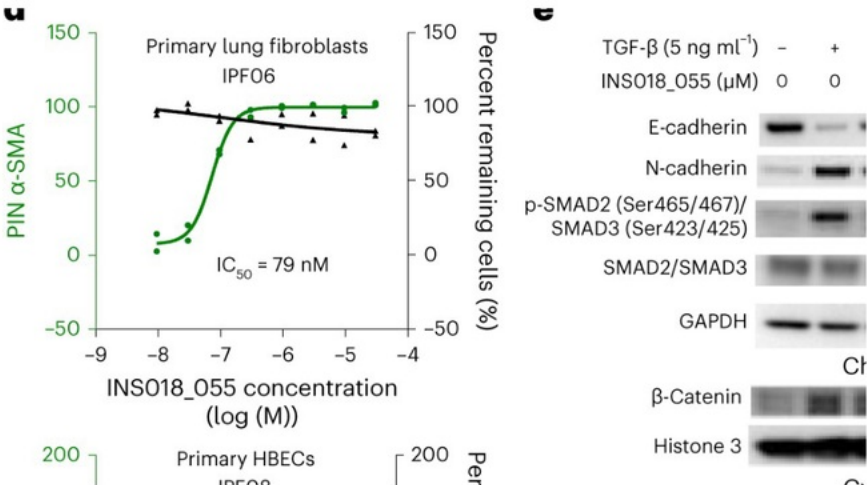
Experimental anthracycline–DNA complex, PDB 1D12.

RCSB PDB 1D12; drug–nucleic-acid recognition context.

29

Check the Predicted Interaction Directly

- A binding experiment addresses whether the proposed interaction occurs.
- It does not by itself establish the entire cellular mechanism.



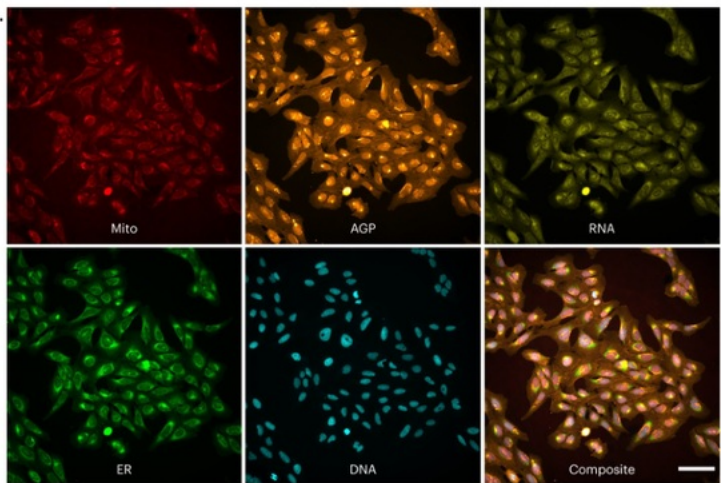
Published TNIK binding evidence; this is a measurement, not a docking score.

Ren et al., Nat. Biotechnol. 43, 63–75 (2025; online 2024).

30

Use Cell Images to Measure the Response

- Images capture changes in cell shape and internal organization.
- AI-assisted image analysis needs controls and a clearly defined biological question.

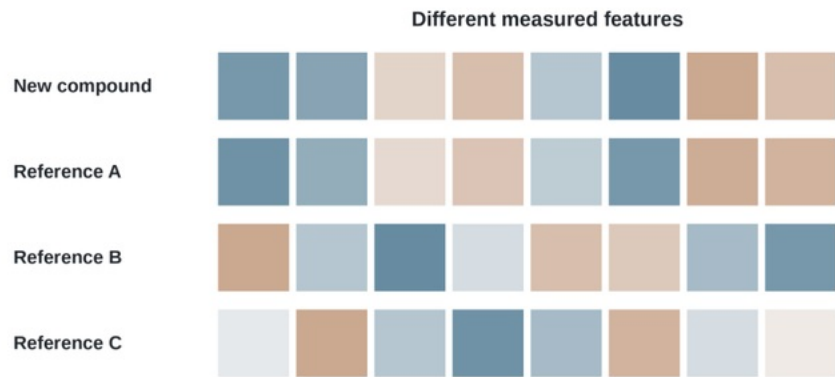


Published Cell Painting example; images are experimental inputs.

Chandrasekaran et al., Nat. Methods 21, 1114–1121 (2024).

Work Backwards from an Unexplained Cell Effect

- Compare the compound's profile with well-characterized references.
- A similar pattern suggests a mechanism to test, not a proven shared target.



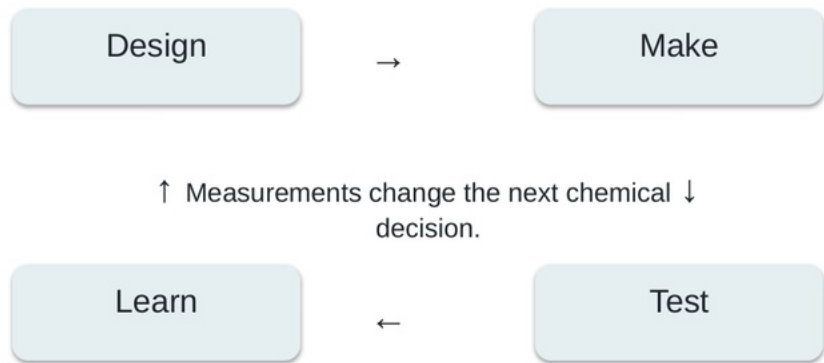
Similar profiles suggest a testable mechanism—not proof of one

Simulated profiles show how reference matching can generate a hypothesis.

Chandrasekaran et al., Nat. Methods 21, 1114–1121 (2024).

Let Experiments Change the Next Design

- Use both successful and unsuccessful compounds to refine the next choice.
- The best next experiment may test a new chemical idea, not the highest score.

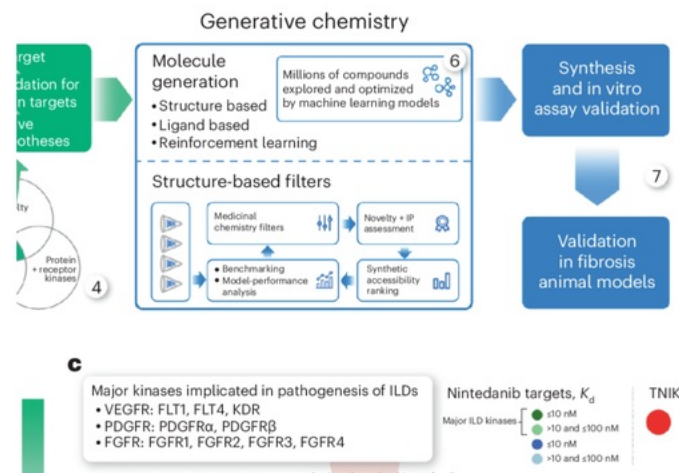


Design → make → test → learn: AI supports the experimental cycle.

Loeffler et al., J. Cheminform. 16, 20 (2024).

Case 1: AI Helped Design TNIK Inhibitors

- After target prioritization, Chemistry42 proposed candidate structures.
- Synthesis and experimental filtering guided candidate selection.

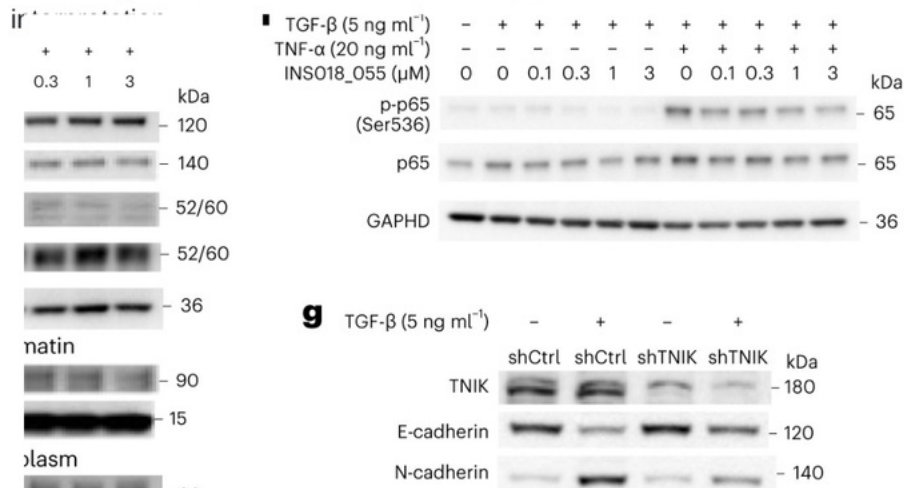


Published structure-based design inputs and filters—not a fully automated discovery.

Ren et al., Nat. Biotechnol. 43, 63–75 (2025; online 2024).

Case 1: Separate Prediction from Measured PD

- The candidate bound TNIK and inhibited the enzyme.
- Cell assays showed fibrosis-related effects; those need mechanistic

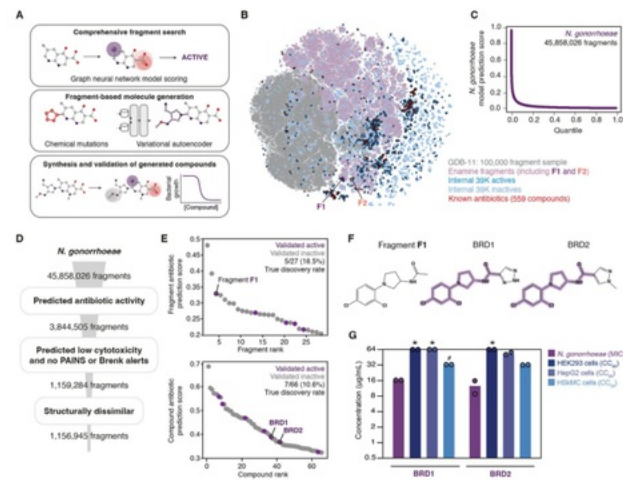


Published cellular assay evidence for INS018_055.

Ren et al., Nat. Biotechnol. 43, 63–75 (2025; online 2024).

Case 2: Generate Antibiotic Candidates

- Expand a selected fragment or propose compounds without a fixed fragment.
- Predicted activity and toxicity guide which molecules are made.

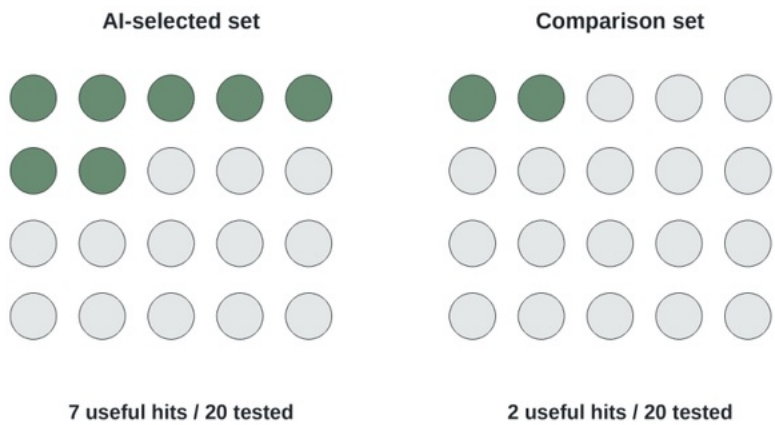


Published generative-antibiotic strategy; candidate generation precedes validation.

Krishnan et al., Cell 188, 5962–5979.e22 (2025).

Measure Whether AI Improved the Decision

- Count useful hits among compounds actually tested.
- Compare with a sensible alternative under comparable conditions.



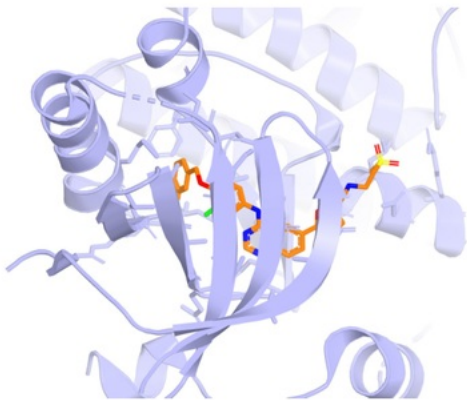
Hypothetical example: twenty experimentally tested compounds per group.

Teaching comparison; no claimed performance for a named model.

43

Exercise: Choose the Next EGFR Experiment

- You have an EGFR structure and a shortlist of candidate inhibitors.
- Which tool addresses the pose—and which assay addresses inhibition?



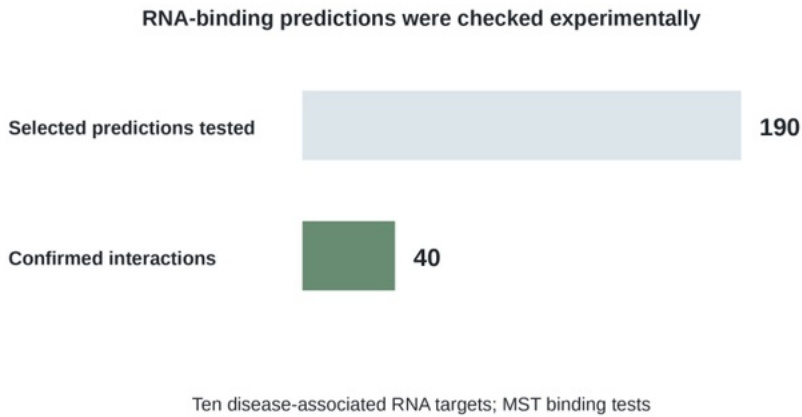
Experimental lapatinib–EGFR complex, PDB 1XKK, as the exercise reference.

Wood et al., Cancer Res. 64, 6652–6659 (2004); PDB 1XKK.

44

Exercise: What Did an RNA Binding Screen Prove?

- SMRTnet nominated interactions across ten disease-associated RNA targets.
- What additional evidence is needed to connect binding to a cellular effect?



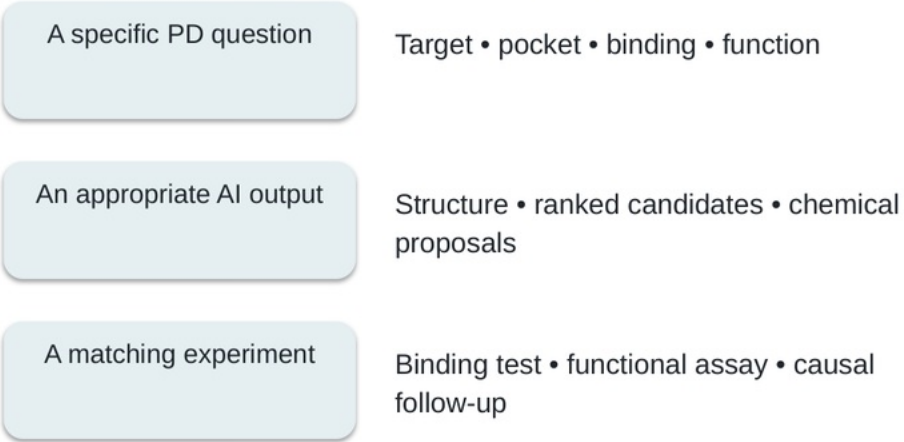
Published counts: 190 predicted interactions tested; 40 confirmed by MST.

Fei et al., Nat. Biotechnol. (2026), doi:10.1038/s41587-025-02942-z.

45

From Target Knowledge to Better PD Decisions

- Define the target interaction or biological effect.
- Choose a suitable AI tool, inspect the chemistry and test the claim.



Chemical knowledge makes AI predictions useful—not automatically true.

Summary of the cited research examples.

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